॥वसुधैव कुटुम्बकम्॥

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 15

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – 3rd Sem
Title of Assignment	Write a Java program to demonstrate exception handling using finally, throw, and throws.
Practice questions	1. Create a Login program that throws an exception for invalid password and uses finally block.

```java
public class LoginDemo {
    public static void main(String[] args) {
        String correctPassword = "Java@123";
        String enteredPassword = "wrongpass";

        try {
            if (!enteredPassword.equals(correctPassword)) {
                throw new IllegalArgumentException("Invalid
password");
            }
            System.out.println("Login successful");
        } catch (IllegalArgumentException e) {
            System.out.println("Login failed: " + e.getMessage());
        } finally {
            System.out.println("Login attempt completed");
        }
    }
}
```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```
public class ATMPinVerificationDemo {
    public static void main(String[] args) {
        int correctPin = 4321;
        int enteredPin = 1234;

        try {
            if (enteredPin != correctPin) {
                throw new IllegalArgumentException("Invalid PIN
entered");
            }
            System.out.println("PIN verified successfully");
        } catch (IllegalArgumentException e) {
            System.out.println("Verification failed: " +
e.getMessage());
        } finally {
            System.out.println("PIN verification process
completed");
        }
    }
}
```

Output

1)

```
\redhat.java\jdt_ws\JAVA_PRACTICALS_eabe1187\bin' 'JAVA_EX
Login failed: Invalid password
Login attempt completed
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

2)

```
cationDemo'
Verification failed: Invalid PIN entered
PIN verification process completed
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion	throw and throws give a program precise control over when and how exceptions are raised and propagated, while finally ensures important cleanup or completion logic always runs, whether an exception occurs or not. Together they make exception handling more predictable and reliable, which is essential for critical operations like login checks and ATM transactions.